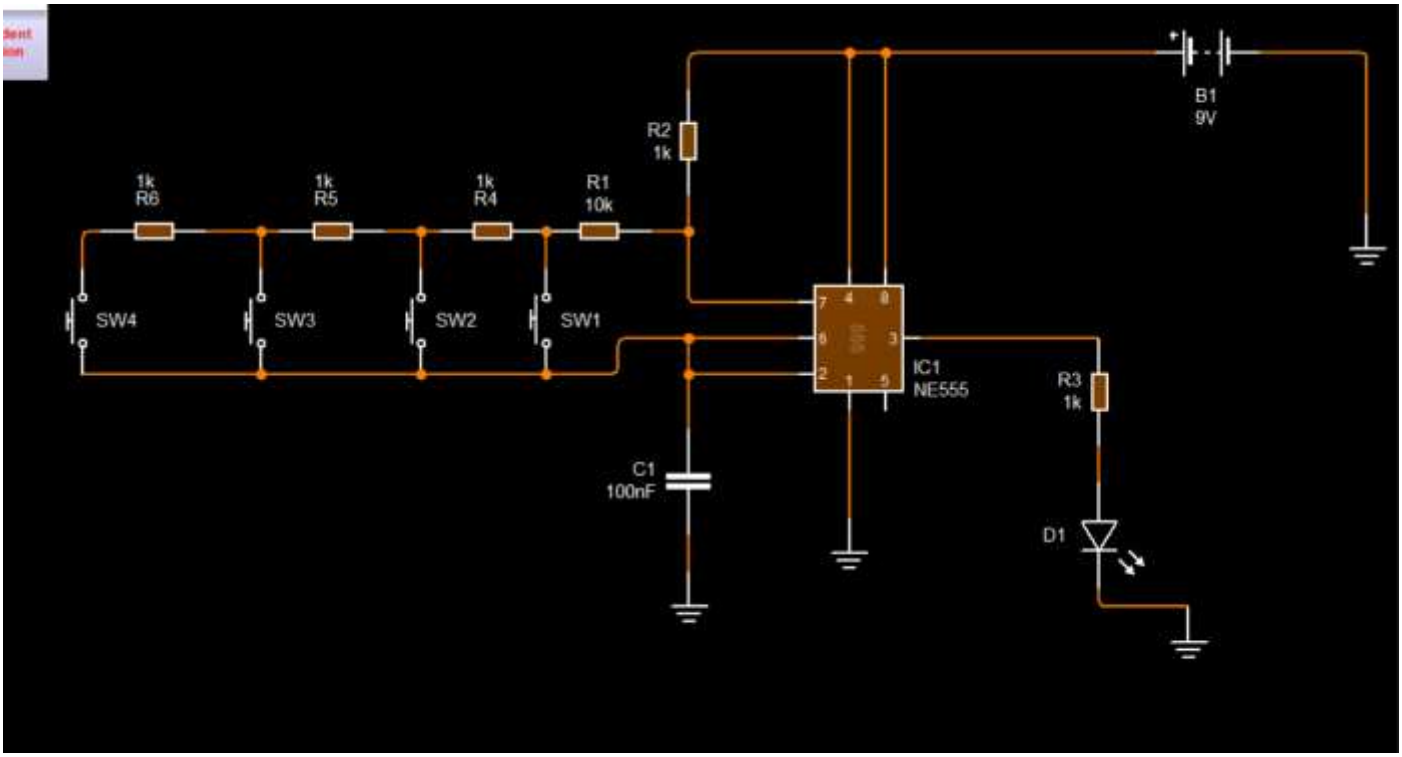
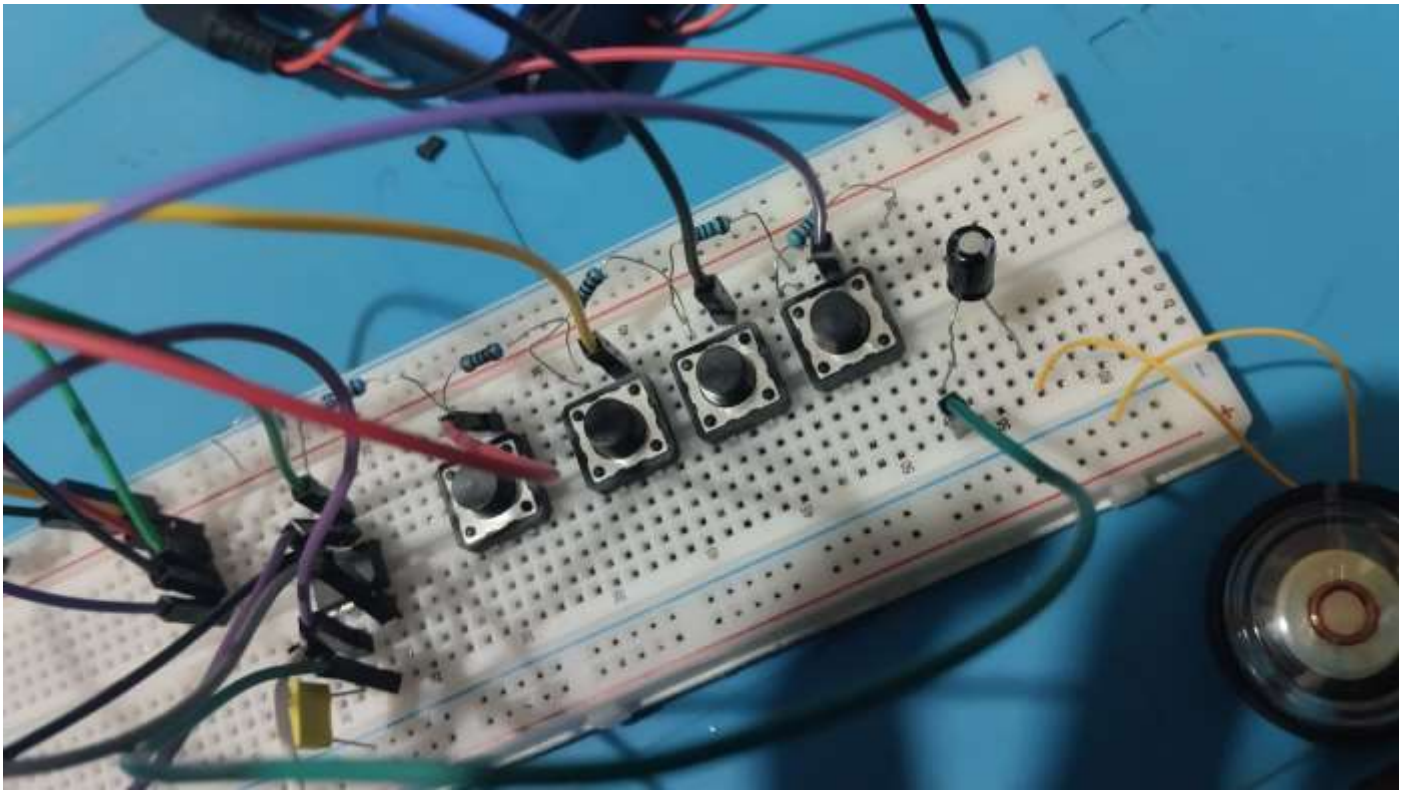
Multi-Tone Sound Generator Using NE555 Timer and Switch-Controlled Network

This project demonstrates a sound generator circuit based on the NE555 timer, configured to produce multiple audio tones using selectable resistor paths.

The circuit uses a combination of push-button switches (SW1–SW4) and resistors (R4–R6) to modify the timing characteristics of the 555 timer. By pressing different switches, the effective resistance in the timing network changes, which directly affects the oscillation frequency.

How It Works:

- The NE555 timer operates in astable mode, generating a square wave signal.
- The frequency of oscillation depends on the RC network (resistors + capacitor).
- Each switch activates a different resistance path → producing a different tone.
- The output signal is sent to a speaker, converting electrical pulses into sound.